Blockchain Integration in 5G Technologies: SDN and Network Slicing

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***Abstract*—The deployment of fifth-generation wireless networks shows promising significant advancements in network capacity, system throughput, and high-quality service. Nevertheless, challenges such as transparency, network privacy, and security vulnerabilities remain to be addressed. Blockchain, a distributed ledger system, offers transparency, immutability, and decentralization, making it a compelling solution for addressing these issues in the context of 5G. By combining blockchain and 5G, new opportunities arise for secure and transparent IoT networks, enhanced network security and privacy, as well as decentralized network management and governance. This paper delves into the application of blockchain in 5G, discussing its benefits and challenges. Interestingly, we concentrate on two key prospective applications, namely software defined networking and Network Slicing. We finally discuss the challenges for widespread adoption of blockchain in the context of 5G and provide insights for future research directions.**

***Index Terms*—5G, Blockchain, Network Slicing, Software Defined Networking, Security, Privacy.**

I. Introduction

The advent of fifth-generation (5G) wireless networks [1] brings a paradigm shift in the telecommunications industry, promising unprecedented advancements in network capabilities. With its remarkable improvements in data transfer rates, ultra-low latency, and massive connectivity, 5G technology holds the potential to revolutionize various sectors and enable a wide range of innovative applications.

With the development of Software-defined Networking (SDN), 5G offers the deployment of services with their network, called Network Slicing. SDN and Network Slicing allow 5G networks to be divided into smaller areas for easy control and maintenance, meeting various strict user requirements.

According to these advantages, 5G networks support a massive number of connections between heterogeneous devices and transfer data at high speed along with low latency. Such an enormous number of distributed and heterogeneous devices raises new transparency challenges and network privacy and security vulnerabilities to 5G networks. Traditional centralized approaches like centralized data centers or centralized cloud providers [2] may struggle with the highly dynamic and diverse 5G ecosystem. For instance, many 5G applications, such as autonomous vehicles, augmented reality, and remote surgery, demand ultra-low latency. Centralized data centers are often geographically distant from end-users, leading to latency issues. Besides, 5G networks require high levels of reliability and redundancy to ensure uninterrupted service. Centralized systems can be vulnerable to single points of failure. Therefore, alternative solutions that offer enhanced security, reduce latency, and offer fault tolerance are considered to address these challenges. To address those challenges, numerous surveys have cited blockchain technology as a viable solution. Blockchain, at its core, is a decentralized and distributed system whose state is publicly verifiable and maintained by thousands of nodes. With transparency, immutability and decentralization, blockchain offers opportunities to deal with the limitations of 5G networks.

The rest of this paper is organized as follows. Section II provides a literature review of existing surveys that mention the integration of blockchain and 5G technologies. Section III provides our survey on the adoption of blockchain in 5G technologies, specifically SDN and Network Slicing. Section IV shows the current challenges and highlights its future research directions. Finally, Section V concludes the paper.

II. Literature Review

There are several surveys that address the usage of Blockchain in Network Slicing [3] and SDN [4].

In Ref. [3], the paper highlights the potential of blockchain technology in automating costly manual processes and administrative negotiations among stakeholders in Network Slicing. It suggests that blockchain can help streamline the deployment of network slices and enable end-to-end control of services and resources. However, the paper does not cover all the potential limitations and challenges associated with Network Slicing itself, such as the dynamic environment in 5G and different requirements for each slice in Network Slicing.

In Ref. [4], Nam et al. provide a comprehensive survey of Blockchain technologies applied to SDN in both security and non-security fields. It discusses significant research challenges and perspectives in Blockchain-based SDN, including scalability, cost and energy consumption. Although the paper acknowledges that while a Blockchain-based SDN system offers benefits such as network management without device modifications and secure information storage, it has not yet shown solutions for challenges related to security and fault tolerance.

Our paper provides a literature review on updated applications of Blockchain in 5G technologies: SDN and Network Slicing. The main contributions of the paper are shown in Fig. 1:

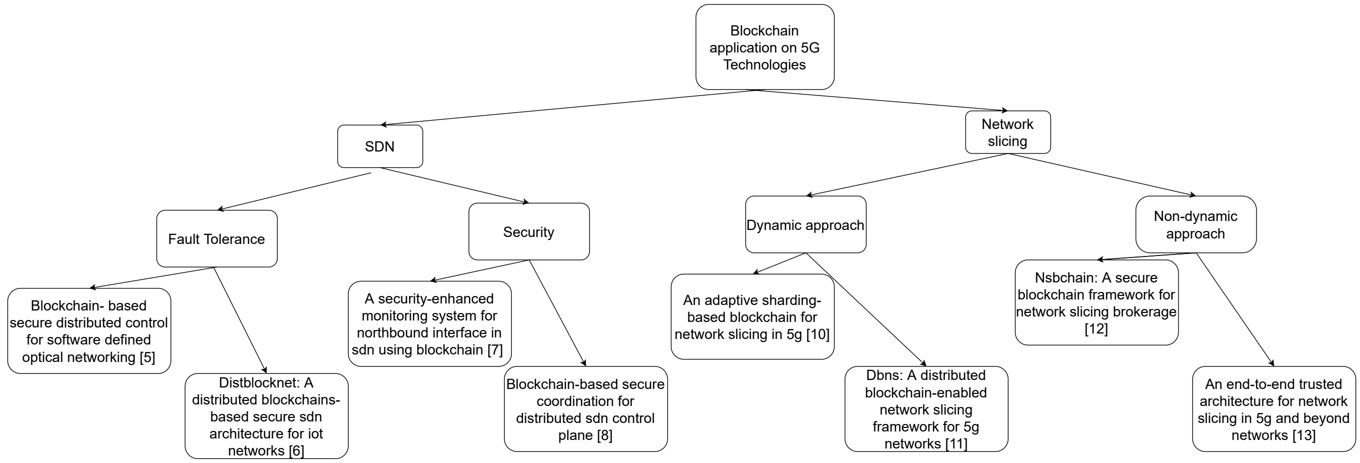

Fig. 1: Blockchain applications in SDN and Network Slicing

- Conduct a comprehensive survey of existing approaches that explore the integration of blockchain technology into Software-Defined Networking (SDN) environments, with a particular emphasis on enhancing security and fault tolerance mechanisms.
- Survey existing approaches to apply blockchain into Network Slicing with a focus on the dynamic and non-dynamic approaches to tackle resource sharing and management issues in Network Slicing.
- Highlight the remaining challenges and the future research directions.

III. BLOCKCHAIN FOR ENABLING 5G TECHNOLOGIES

A. Blockchain for SDN

SDN brought a groundbreaking idea to facilitate network system management by decoupling and abstracting the control plane and data plane of traditional networks. This design architecture provides flexibility and programmability to SDN and helps network administrators manage easily, programme, and troubleshoot network problems. In the 5G context, SDN makes the connectivity services provided by 5G networks programmable, where traffic flows can be dynamically controlled to achieve maximum performance benefits. However, despite the obvious advantages, there remain some challenges that hold back the adoption of SDN in 5G.

Although considering the future of Internet architecture, SDN still has its weaknesses in security. The centralized SDN controller also brings the single point of failure attack on the control layer, which can cause controllers, routers, and switches to be maliciously compromised, generating and causing loss of flow table information. Whereas distributed controllers might be vulnerable to insider attacks.

The goal of achieving a fault-tolerant control system with high efficiency in SDN is addressed in the research conducted by [5]. The data plane assumes the role of facilitating the fundamental data forwarding function, orchestrated through the OpenFlow protocol. Within the control plane, a distributed network of controllers is interconnected through blockchain technology, operating across distinct control domains. At the software layer, every controller within this control plane is equipped with an identically distributed ledger managed by the consensus plane. Smart contracts draw upon the consistent data within this distributed ledger to offer tailored network functionalities. The consensus plane undertakes multi-controller consensus for pending service processes, incorporating the outcomes into a block data structure within the distributed ledger. Simultaneously, the contract plane houses smart contracts for the automated execution of network functions. This blockchain-based approach presents a viable solution for tackling various security concerns, including fault tolerance facilitated by blockchain consensus mechanisms and data consistency achieved through a distributed ledger, eliminating the need for intermediary entities.

The paper [6] proposes DistBlockNet, a new architecture based on Blockchain technology mainly deployed in a distributed SDN to enhance the security, flexibility, and scalability of the IoT network. The system consists of several components, including OrchApp and Shelter modules, designed to provide security mechanisms and protect against attacks. OrchApp is responsible for programming characterized fortifications at the application layer, deploying security policies, and threat intelligence for access control and data protection. Shelter module protects the SDN environment against insider attacks with a flow control analyzer and a packet migration component. The framework also includes an OpenFlow-based framework named OpenSec, which helps maintain pre-defined security policies. DistBlockNet utilizes replication techniques to avoid single points of failure, but it may cause Byzantine Failure. Using blockchains in DistBlockNet brings benefits like reasonable incentive models, smart contract operations, and scalability.

In Ref. [7], the authors propose a security-enhanced monitoring system called BlockAS for the Northbound Interface in Software-Defined Networking (SDN) using blockchain technology. It addresses the need for access control, authentication, and authorization mechanisms between controllers and applications at the Northbound Interface. The system provides authentication, authorization, and accounting (AAA) functionalities to protect SDN controllers from malicious applications. It ensures the immutability of the database, decentralization, and secure access to critical controller resources from applications.

In Ref. [8], the authors design and build blockchain-coordinating controllers (BCC) to secure control communications of the SD-WAN controller network formed by distributed controllers spread across multiple domains. The authors validate BCC's effectiveness and efficiency by prototyping it using Ethereum and smart contracts on CloudLab. It demonstrates that BCC achieves distributed consensus at a sub-second level for certificate/key distribution and network-wide control communication synchronization. The system also protects against security threats in the control plane, even when up to n controllers' networking credentials are compromised.

B. Blockchain for Network Slicing

One of 5G's critical features is Network Slicing. This allows multiple virtual networks to exist on a single physical network layer, and network operators can divide their resources into different logical networks. Each network slice has unique capabilities to meet the specific requirements of various users and applications, like low-latency support for autonomous vehicles or high-bandwidth support for video streaming. Network Slicing enables the full potential of 5G networks by providing flexible and efficient resource allocation for various use cases. However, Network Slicing introduces new security, privacy and trust concerns, such as resource allocation and management, and so on. Failure to properly allocate and manage resources can lead to congestion, degraded performance, and reduced user satisfaction.

Blockchain can help ensure resource allocation and management transparency and prevent resource hoarding and misuse. For example, we can use blockchain as a distributed ledger to track the allocation and usage of resources in real time, which can help network operators identify available resources, allocate resources to different users and services based on their needs, and monitor the resource usage of each party in the network. As a result, if a malicious party monopolizes or misuses resources, the network administrator can quickly identify it and take appropriate action. Despite these advantages, one of the significant drawbacks of blockchain is its limited throughput. For example, Bitcoin provides 7 TPS and Tendermint-based blockchains like Cosmos around 400 TPS [9]. These throughputs are still a far cry from what could be generated by a large-scale network like 5G. As a result, blockchain application to large-scale networks such as 5G faces many problems.

To apply blockchain to Network Slicing while meeting the requirement of network throughput, the paper [10] proposes an adaptive sharding-based blockchain. The system has several network slices for different uses, such as connected vehicles, personal computers, and surveillance cameras. Each slice has an independent blockchain network called a "shard", which maintains several blockchain nodes. Each shard works independently and is tailored for a specific requirement of a slice, such as high safety for surveillance cameras or high bandwidth and throughput for vehicles. Therefore, the blockchain must have high throughput and a flexible policy to meet various application demands. Hence, the paper introduces using Deep Reinforcement Learning (DRL) technique to dynamically adjust the sharding parameters of each shard, such as consensus algorithm and block size. This adaptive sharding-based blockchain approach maintains a balance between performance and security, allowing it to tackle the resource sharing issue in Network Slicing environment.

In Ref. [11], the paper proposes a distributed blockchain-enabled Network Slicing (DBNS) framework that allows service and resource providers to dynamically lease resources for high-performance end-to-end services. The framework includes a global service provisioning (GSP) component that provides admission control for incoming service requests and dynamic resource assignment through a blockchain-based bidding system. The goal of the framework is to enhance users' experience with diverse services and reduce providers' capital and operational expenditures.

In [12], the author presents NSBchain, a secure blockchain framework for Network Slicing brokerage that addresses the needs of new business models in the telecom market. It leverages blockchain technology to enable the allocation and redistribution of network resources in a secure, automated, and scalable manner. It defines an entity called the Intermediate Broker (IB) that allows Infrastructure Providers (InPs) to allocate network resources to IBs through smart contracts. IBs can then assign and redistribute these resources among tenants, which refers to a business entity or organization that utilizes network resources, ensuring efficient resource utilization. The framework utilizes smart contracts and distributed consensus algorithms to guarantee secure resource exchange and enable the system to evolve autonomously without centralized authorities. NSBchain supports the network slice resource brokerage process in an end-to-end fashion, accommodating heterogeneous tenants' requirements in a multi-tenant multi-domain scenario. The use of blockchain technology ensures data transparency, as the content of the blockchain database is broadcast and updated in a decentralized manner.

In Ref. [13], the authors propose a Blockchain-based trust architecture to create and negotiate the deployment of end-to-end network slices, manage service level agreements (SLAs), and ensure security and anonymous transactions. The paper introduce new stakeholders in the Network Slicing model, including Verticals or network slice tenants, Network Slice Providers, and Resource Providers. The proposed framework successfully automates the process of managing SLAs among the involved actors, removing the dependency on a third-party player for handling the billing process. It also highlights the need for future extensions to address the challenge of trust monitoring, as it is crucial for improving the trust framework.

IV. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

This section highlights and discusses challenges that may hinder the massive adoption and utilization of blockchain in

5G.

A. Scalability

The desired end-to-end latency in 5G networks is set to be below one millisecond for both payload and carried data. This exacting demand necessitates the execution of configuration and setup transactions at a notably high throughput pace. Different 5G applications have varying transaction speed requirements, and although it is challenging to provide precise figures, blockchains should be capable of supporting thousands of transactions per second (TPS) to effectively serve 5G applications. Unfortunately, public blockchain networks typically offer throughputs one or two orders of magnitude lower than the required amount (e.g., Bitcoin provides 7 TPS and Cosmos around 400 TPS). These throughput limitations primarily stem from the consensus protocols used to validate transactions, block size/rate limits, and the absence of transaction validation parallelization. Consequently, scalability solutions, such as sharding, block size/rate increases, and advancements in consensus algorithms, are currently under research to enhance the throughput of blockchain networks.

B. Security and Latency

Besides scalability, balancing security with network latency is also a critical challenge. Blockchain technology introduces an additional layer of security, but this can potentially lead to increased latency in network operations. Research is needed to devise innovative approaches that mitigate latency while ensuring the integrity and security of network transactions.

C. Adaptability

Most current sharding-based blockchains run with fixed parameters (block size, consensus algorithm, etc.), which can not be adaptable to the dynamic environment in 5G. For example, in Network Slicing, each slice may have different requirements, such as a slice used for surveillance cameras requiring a high level of security or a slice used in autonomous vehicles that demands low latency and data loss. Hence, blockchain needs to have dynamic policies allowing it to change the parameters based on the environment.

D. Consensus Mechanisms

Traditional blockchain consensus mechanisms may not be optimized for the unique requirements of SDN and Network Slicing. For example, traditional PoS mechanisms, which select validators based on the amount of tokens they hold, may not be well-optimized for the dynamic and real-time nature of SDN and Network Slicing. In such scenarios, validators' stakes in tokens might not inherently reflect their ability to make rapid and accurate decisions regarding network resource allocation. Developing consensus algorithms tailored to the dynamic and diverse nature of these environments is in need These algorithms should consider factors like the huge number of connected devices in 5G, resource allocation, resource management, and fault tolerance specific to SDN and Network Slicing.

E. Interoperability and Standardization

Ensuring interoperability and adherence to standards is a complex challenge that goes beyond individual blockchain implementations. In 5G networks, which include a variety of technologies and devices, it becomes crucial to integrate blockchain solutions seamlessly. To tackle this challenge effectively, research efforts should prioritize the development of standardized protocols and interfaces that enable different components and systems to work together harmoniously.

V. CONCLUSION

In this paper, we present a survey of the adoption of blockchain in 5G technologies, particularly in SDN and Network Slicing. Blockchain integration can provide 5G benefits such as manageable, programmable, flexible and efficient resource allocation for various use cases. Furthermore, we identified several open research challenges that need to be addressed in order to utilize and integrate blockchain in 5G networks efficiently.

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